

AHMED FERGANEY

Data Analyst | Business Intelligence & Analytics Engineering

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PROFILE

Data Analyst and BI professional with hands-on manufacturing experience transforming production, quality, loss, and job-change data into decision-ready insights. Skilled in Power BI, DAX, Power Query, SQL, Python, dimensional modeling, and SSIS, with experience building validated star-schema models, KPI frameworks, and executive/engineering dashboards. Combines analytics engineering with manufacturing domain knowledge to quantify losses, identify root causes, and support data-driven improvement decisions.

EDUCATION

B.Sc. in Mechanical Engineering | Banha University | 2016 – 2022

- Academic Standing: Ranked 5th out of 120 in the Mechanical Engineering Department

CERTIFICATIONS

- AI Engineer Core Track (LLM Engineering, RAG, QLoRA, AI Agents) — Udemy (2026, in progress)
- Prompt Engineering & RAG — Eng. Abu Bakr Soliman (2026)
- FastAPI and Cloud Computing — Udemy (2025)
- Machine Learning & Deep Learning Courses— Stanford University (2024)
- Databases, SSIS, SSAS, SSRS, CCNA, and Docker — Udemy (2024)
- Statistics, Linear Algebra & Numerical Optimization — Dr. Hatem Elattar (2024)
- Embedded Systems and Linux Diploma — Amit Learning (2023)

TECHNICAL SKILLS

Data Analysis & BI: Power BI (Data Modeling, DAX, Power Query, RLS), SQL Server, Data Cleaning, Data Visualization, KPI/Metric Design, Trend & Pattern Analysis, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly.

Data Engineering & Modeling: Dimensional Modeling (Star/Snowflake Schema), SSIS, SSRS, SSAS, Data Warehousing, Microsoft SQL Server, MongoDB, Query Optimization, Database Design

Mathematics & Statistics: Statistics, Linear Algebra, Numerical Optimization

AI & Machine Learning: Machine Learning, Deep Learning, LLM Engineering, AI Agents, Prompt Engineering, RAG, PyTorch, TensorFlow

Cloud & DevOps: Docker, Kubernetes, CI/CD Automation, Observability (Prometheus, Grafana)

Programming: Python, C, SQL, Bash, OOP, Data Structures & Algorithms

Tools: Power BI, MS Office, Git/GitHub, Postman, Jira, Trello

PROFESSIONAL EXPERIENCE

Product Engineer | Kandil Glass | 10th of Ramadan City, Egypt 06/2026 – Present

- Built an end-to-end manufacturing analytics solution: modeled production data into a star-schema data warehouse with an SSIS ETL pipeline and developed Power BI/DAX dashboards to track KPIs, uncover trends, and support data-driven decisions.
- Developed a 5-page Manufacturing Performance & Loss Analytics dashboard integrating production, process-loss, quality, rework, and job-change data to analyze FPY, defect/hold rates, production gaps, downtime reasons, and T1/T2 changeover performance.
- Extended loss attribution from Line → Section → Cavity and implemented capacity-weighted equivalent loss hours and estimated lost-unit KPIs, enabling engineering teams to prioritize the highest-impact assets.

Operational Excellence Engineer | Midea Electric | 6th of October City, Egypt 01/2025 – 05/2026

- Analyzed operational data and built Power BI dashboards to track KPIs, downtime, and productivity trends across production lines.

Process Quality Engineer | Methode Electronics | Nasr City, Egypt 12/2023 – 12/2024

- Collected and analyzed quality data to identify defect trends, building Excel reports for cross-functional teams.

Continuous Improvement Engineer | Jac | Badr City, Egypt 03/2023 – 11/2023

- Applied data-driven analysis of production KPIs and process metrics to optimize manufacturing performance, improve OEE, and reduce defects.

LANGUAGES & ADDITIONAL SKILLS

Languages: Arabic (Native), English (Professional), German (Basic)

Leadership: Cross-functional Team Leadership, Agile/Scrum Methodologies, Technical Mentoring, Problem-Solving Under Constraints

PROJECTS

Manufacturing Performance & Loss Analytics Power BI Platform | [GitHub](#)

- Built a 5-page Power BI analytics solution covering FPY/pack efficiency, production gaps, process losses, product/order performance, quality/rework, job-change efficiency, and downtime-reason analysis.
- Designed a dimensional semantic model across production, process-loss, quality, rework, and job-change facts, supported by governed Excel inputs, stable event IDs, data validation, and reconciliation rules.
- Extended loss analysis from line level to 13-section × up-to-3-cavity asset granularity and implemented capacity-weighted equivalent loss hours and estimated lost-unit calculations to prioritize high-impact sections and cavities.
- Developed DAX measures for KPI ratios, T1/T2 target variance, ranking, dynamic conditional formatting, Top-N analysis, and context-aware executive reporting.

AdventureWorks Data Warehouse & Power BI Analytics | [GitHub](#)

- Designed a Kimball star-schema data warehouse (FactSales + 4 dimensions) from the AdventureWorks OLTP database.
- Built a full SSIS ETL package (Control Flow + Data Flow) to extract, transform, and load data; deployed to the SSIS Catalog and scheduled daily via SQL Server Agent.
- Developed Power BI dashboards with DAX measures (Revenue, YTD, YoY growth) covering sales trends, product performance, and territory analysis.

Customer Segmentation using Machine Learning | [GitHub](#)

- Segmented an automotive dataset using KMeans, Random Forest, and Linear Regression; applied feature engineering, outlier detection (DBSCAN), and PCA for actionable customer insights.

Transfer Learning for Image Classification | [GitHub](#)

- Implemented transfer learning pipelines using pre-trained models (ResNet, VGG, EfficientNet) with fine-tuning strategies, data augmentation, and hyperparameter optimization to achieve production-ready classification accuracy

Bilingual SMS Spam Classification (NLP) | [GitHub](#)

- Built a bilingual (Arabic/English) spam detection system using NLTK and scikit-learn with multiple vectorization methods and a validated Random Forest model.

Async Document Intelligence RAG System | [GitHub](#)

- Built a distributed FastAPI/Celery/RabbitMQ backend with a retrieval + reasoning pipeline (LLMs + vector search); optimized API latency to <150ms with async I/O and Redis caching.

Fine-Tuning LLMs for Arabic News Processing | [GitHub](#)

- Fine-tuned Qwen2.5-1.5B-Instruct with LoRA for structured entity extraction and translation; deployed a vLLM inference server with 80% GPU utilization.